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CHE 312 — Past Questions & Materials
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Course: CHE 312
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 300

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Paper 1 of 1 — CHE 312 Transport Phenomena

Level 300 • First Semester • PastQuestion

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COLLEGE OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF CHEMICAL ENGINEERING

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FIRST SEMESTER EXAMINATION, 2024/2025 ACADEMIC SESSION

COURSE TITLE: TRANSPORT PHENOMENA I

COURSE CODE: CHE 312 CREDIT UNIT: 3 UNIT

TIME ALLOWED: 3 HOURS

INSTRUCTION: There are seven (7) Questions; answer five questions in all, with at least two questions from each section.

SECTION A

1. (a) Using the analogous relationships in transport phenomena,

(i) Identify the three transport coefficients and their units,

(ii) Mathematically express Newton's, Fourier's, and Fick's laws, highlighting their shared proportionality between flux and gradient,

(iii) Compare the role of each transport coefficient in momentum, heat, and mass transfer.

(iv) Propose one Chemical Engineering application where the three phenomena interact.

(b) i. Using Fick's Law of Diffusion, calculate the steady-state flux of acetic acid (A) across a 2mm stagnant water film (B) at 17°C, given concentrations of 9 wt% and 3 wt% on either side, solution densities of 1012 kg/m³ and 1003 kg/m³ respectively, and a diffusivity of $0.95 \times 10^{-7} \text{ m}^2/\text{s}$

(ii). How would the diffusion rate change if the film thickness were doubled?

2. (a) Analyze the key differences between diffusive and convective mass transfer by comparing their driving mechanisms, governing equations, and practical applications.

(b) Air at 20 °C ($p = 1.205 \text{ kg/m}^3$, $\nu = 15.06 \times 10^{-6} \text{ m}^2/\text{s}$, $D = 4.166 \times 10^{-6} \text{ m}^2/\text{s}$) flows over a tray"

. f

(length = 320mm, width = 420 mm), full of water with a velocity of 2.8 m/s. The total pressure of moving air is 1 atm, and the partial pressure of water present in the air is 0.0068 bar. If the temperature on the water surface is 150 °C, calculate the evaporation rate of water.

3. (a) LifeSaver Pharmaceutical Plc. is considering a Polyethylene film 0.04 cm thickness for use in packaging its newly formulated drug (Tracunizole) at 30 °C. The partial pressure of oxygen

outside the package is 0.21 atm, and inside the package is 0.015 atm. Assuming steady state conditions, and that the resistance to diffusion outside and inside the film is negligible compared to the film's resistance.

- (i) What is the diffusion flux of Oxygen ($\text{kgmol/m}^2\text{s}$) through a m^2 cross-section
- (ii). How many kilograms of Oxygen pass per day through a square meter cross-section of the polyethylene film?
- (iii). Given the same temperature and pressure conditions, would you advise the pharmaceutical firm to use a nylon film of the same thickness as a more suitable packaging for its new drug? Justify your answer. Use the Permeability data from Table 1

Hint: P_a (Permeability of polyethylene) = $4.17 \times 10^{-12} \text{ 7202s}$.

S.m? atm/m

- (b). The diffusion coefficient of oxygen is approximately:

$e 1.8 \times 10^{-5} \text{ m}^2/\text{s}$ in air

$\ll 2.1 \times 10^{-9} \text{ m}^2/\text{s}$ in water

$e 3.0 \times 10^{-7} \text{ m}^2/\text{s}$ in rubber

Explain these differences in magnitude.

- 4. (a) From the data provided, calculate the diffusion of CO₂ in air at a temperature of 25 °C and atmospheric pressure using the Chapman-Enskog Equation for Gas Diffusivity

CO₂ (A): $\sigma = 3.996 \text{ \AA}$; $\epsilon = 190\text{K}$, $M_A = 44$

Air (B): $\sigma = 3.167 \text{ \AA}$; $\epsilon = 97\text{K}$, $M_B = 29$

may be used to estimate the collision integral

The table below

kT/ϵ 2.00 2.10 2.20 2.30 |

1.075 1.057 1.041 1.026 |

(b)

- (i). Differentiate between the three modes of heat transfer
werGen Utilities, you must evaluate a 45 mm-thick copper heat

- (ii). As a process engineer at Po
 $\epsilon 350^\circ\text{C}$ (exhaust side) and 50°C

$^\circ\text{C}$) with face temperatures 0

exchanger plate ($k=370 \text{ W/m}$

and total heat transfer for a 12m^2 plate

(coolant side). Calculate the steady-state heat flux

SECTION B

tion, with a density of 3.57 slug/ft^3) from an open

gh a 3.068-in ID suction. The pump discharges

d tank. The end of the discharge line is

1. A pump draws 261.6 L/s of a liquid solution from a storage feed tank of large cross-sectional area through its flow through a 0.17225 ft — ID line to an open overhead tank 50 ft above the level of the liquid in the feed tank. The friction losses in the piping system is 10 ft — 16 £/lb mass. What pressure must the pump develop, and what horsepower must the pump have if its efficiency is 65%? The flow is turbulent.
- 0.2 gpm through a horizontal nozzle. The nozzle is attached at the upstream, and the frictional forces are considered negligible. The density of the water is 1000 kg/m³.
2. Oil flows through a 1000-foot cast-iron pipe with a 10-inch diameter at a rate of 157.73 L/s. Given that: Oil density: 1.65 slug/ft³, Kinetic Viscosity: 1.20 ft²/s, Pump efficiency: 0.88. Determine:
 - (a) The Reynolds number
 - (ii). What is the power to be supplied by a pump to the fluid?
 - (iii) Fluid the Brake horsepower of the pump
- (b) A fluid with a density of 1000 kg/m³ and a dynamic viscosity of 0.001 Pa.s; flows through a pipe of diameter $D = 0.05$ m at an average velocity of $V = 2$ m/s. Determine the Reynolds number.
- 3.(a) Water at 20 °C flows through a 11811-inch-long steel pipe with a diameter of 7.874 in and absolute roughness of 0.0001476 ft. The head loss is known to be 204.7245 in. Find the volumetric flow rate of water in the pipe.
- (b). Water with a density of 998 kg/m³ is flowing at a steady mass flow rate through a uniform diameter pipe. The entrance pressure of the fluid is 68.9 kN/m² absolute in the pipe, which connects to a pump that supplies 155.4 J/kg of fluid flowing in the pipe. The exit section of the pipe is 3.05 m higher than the entrance, and the exit pressure is 137.8 kN/m² absolute. The Reynolds number in the pipe is above 4000 in the system. Determine the frictional loss in the pipe system.

Table 1: Diffusivities and Permeability in Solids

Dy, :

Diffusion Solubility, S Permeability, P,

Solute Coefficient m solute([STP]) · [Eggetr »

Solid (B) TK) Eevee) me? solid -atm sm? -atmfm

H, — Vulcanized 298 0,85(107%) 0.040 -0,342(107 5)

robber ; . ,

ion as 298 0.21107) 0.070 0.152(107 #°)

N, 298 0,15(10"%) 0.035 0.054(107 7%)

co, 298° O11G07%) . 0.90 1.01(107?)
 H, — Vuleanized = 290010307) 0.051 an
 neoprene . “
 300 -0.180(10"%) 0.053 .
 Hy -Polysthylene 298 6.53(107}4)
 oO, 303 417(10732)
 N, 303° 1.52(107 #7}
 O,. Nylon 303 . 0.029(107 14)
 RN, "303 * 0.0152(107 27)
 Air English. 298 0.15-0.68x 10-*
 leather i :
 H,O Wax 306 - ~ 0.16107 19)
 H;O Cellophane 31 0.91-1.82(10- 34)
 He Pyrex glass 293 ; 4.86(10—1%)
 = 973° . : 20.1107 1)
 He SiO, . 293 2.4~-5,5(107 4s 0.0! .
 H, Fe 293 2:\$9(107 5)
 AL Cu. 293 1.3(10734
 “1 btu = 778.17 Tb; = 1055.06 5
 1 hp =-550 ft -lbg/s = 0.7457 kW
 1 ft-lb(lb,, = 29890 Skg
 1J=1N-mel kg-m?/s?
 on
 List of Formulas
 ina)
 (it)
 Gi)
 (iv)
 | Ww) Day = OGURES —
 Pos
 where, Dag = ‘Tusivity
 spestes ch, can",
 T=A
 = Collision diameter iA (A
 ‘oltision ntewral, and
 vaycights of gas species 4 and # respectively.
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